# PRAKTIKUM PENGUAT OPERASIONAL
## (OP - Amp)

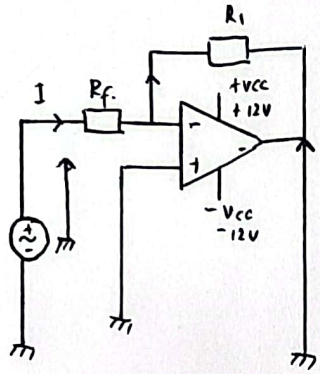
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Kelas : 2A1 TTE

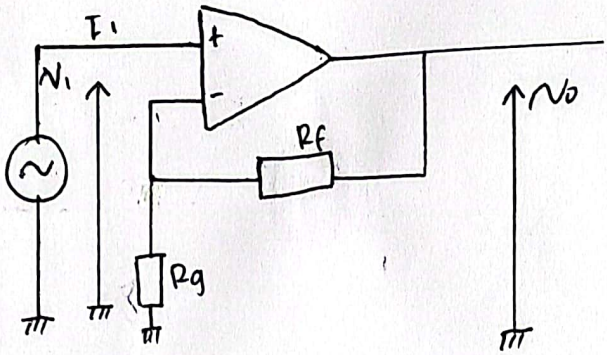
Tanggal
14 September 2023

## A. GAMBAR RANGKAIAN

1. Rangkaian Penguat Inverting




2. Rangkaian Penguat Non Inverting




## B. Perhitungan Teori

① Penguat Inverting:

$$Av = \frac{-Rf}{Rin} = \frac{-1 \cdot 10^3 \Omega}{10 \cdot 10^3 \Omega} = -10$$

$$Vout = \left(-\frac{RF}{Rin}\right) Vi = -\left(\frac{10 \cdot 10^3}{1 \cdot 10^3}\right) 50\ mV$$

$$= -0,5\ V$$

② Penguat non-Inverting

$$Av = \frac{-Rf}{Rg} + 1 = \frac{1 \cdot 10^3 \Omega}{10 \cdot 10^3 \Omega} + 1 = 11$$

$$Vout = \left(\frac{10 \cdot 10^3}{10^3} + 1\right) 50\ mV$$

$$= 0,55\ V$$

## C. Hasil Praktikum:

| | Vo | Vi |
|---|---|---|
| Penguat Inverting | 495 mVpp | 50 mVpp |
| Penguat non-Inverting | 530 mVpp | 50 mVpp |

- Hasil Perhitungan:

-> Penguat Inverting

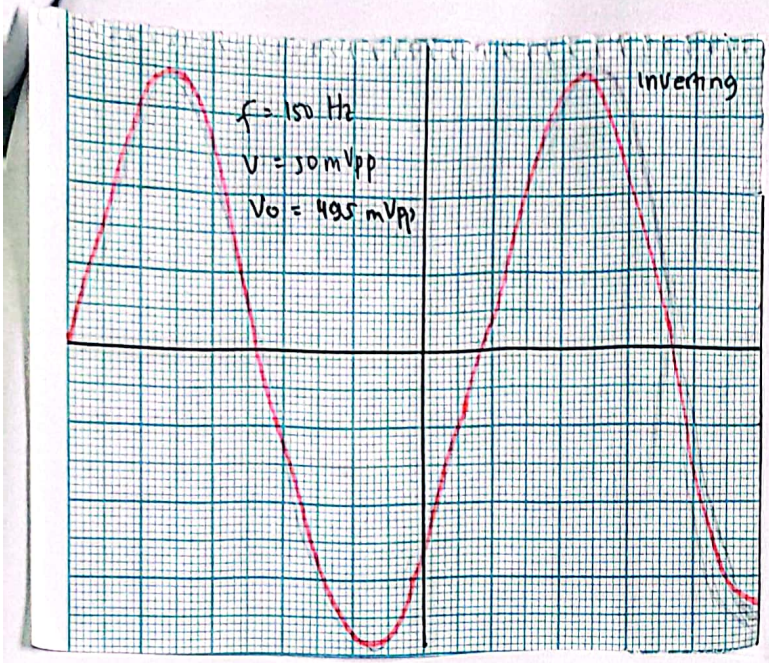
$$Av = \frac{Vo}{Vi} = \frac{495\ mVpp}{50\ mVpp}$$

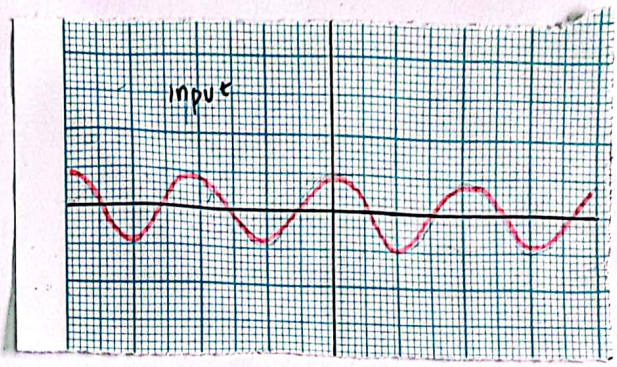
$$= 9,9$$

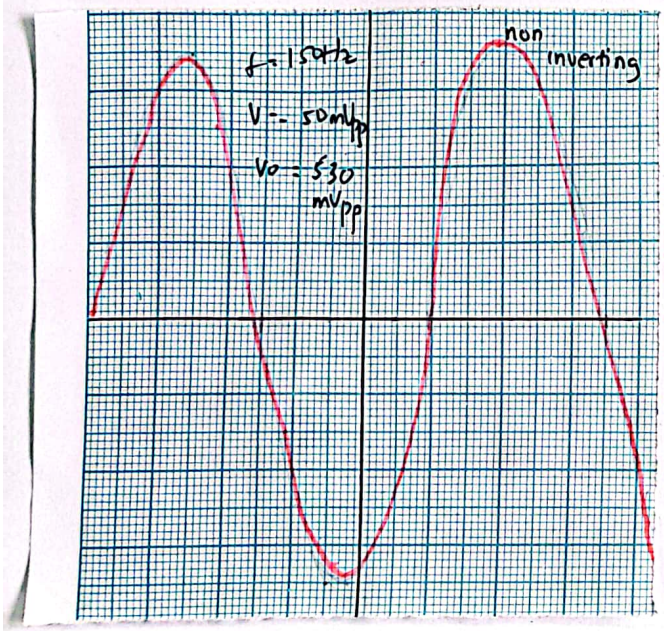
-> Penguat non-Inverting

$$Av = \frac{Vo}{Vi} = \frac{530\ mVpp}{50\ mVpp}$$

$$= 10,6$$


inverting
f = 150 Hz
V = 50 mVpp
Vo = 495 mVpp


input


non inverting
f = 150 Hz
V = 50 mVpp
Vo = 530 mVpp